Conjectures in Grothendieck's working notes

Dated first appearances, open questions, and what can be formalised

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Abstract

The University of Montpellier holds Alexander Grothendieck's mathematical working notes, about 16 000 digitised pages. A transcription project begun in August 2026 has read more than 11 000 of them and restated 164 folders in current notation. This note reports what that reading shows about conjectures. Seven well-known conjectures appear in the notes, some earlier than the texts usually cited for them: the standard conjectures (1965), the de Rham–Betti form of the period conjecture (1965), the generalised Hodge conjecture in ℓ-adic form (1964–about 1971), the p-curvature conjecture (1970s), the homotopy hypothesis (1975), the section conjecture, and the programme behind the Grothendieck–Teichmüller group. The notes also pose questions that have no printed source. For each item we give the place and date in the fonds, the published source to cite, its status, whether it is already in the *Formal Conjectures* repository, and whether current Mathlib can state it. Everything rests on transcriptions that no person has yet read in full against the manuscripts; Section 6 says what follows from that.

1 The corpus and how it was read

The fonds is the deposit of Grothendieck's mathematical notes at Montpellier, catalogued by folder (« cote ») and published as facsimiles at `https://grothendieck.umontpellier.fr`. The project at `https://grothendieck.commutator.io` works on it in four layers.

- **Transcription.** Each batch of twenty pages is transcribed into LaTeX with a critical apparatus (illegible, doubtful, struck, added), by Anthropic's Claude models (Opus 5 and Opus 5.5) under written editorial rules. More than 11 000 pages are done.
- **Modernised readings.** 160 folders are restated whole, in French, in current notation, with footnotes marking every departure from the page. A reading is an interpretation; where it disagrees with the transcription, the transcription is the record.
- **Findings.** A register of 458 entries lists statements of the folders that may not be in the literature, each written as a claim about the literature, with what was searched and what would settle it. Its status vocabulary is used below: *matched* (found in print), *candidate* (searched, not found), *unsearched*, *refuted*.
- **Lean.** An appendix proves in Lean 4, on Mathlib, 2 052 theorems drawn from the readings, using only Lean's standard axioms (`https://grothendieck.commutator.io/article/grothendieck-lean.pdf`). Lean checks the proofs, not that a statement is faithful to the page.

References of the form "shelfmark N, p. k" below point to the reading view of the folder, which shows the transcription beside Montpellier's facsimile.

2 Seven conjectures and where they first appear

Table 1 sums up this section. The dates are those of the leaves or of Montpellier's inventory, whose brackets mark a date the archivists inferred.

Conjecture	In the fonds	Published source to cite	Mathlib today
Standard conjectures	14, p. 44: letter to Serre, 27 Aug. 1965; 16 (1966–67)	Grothendieck, Bombay 1968 (1969); Kleiman 1968, 1994	no Weil cohomology
Period conjecture (de Rham–Betti)	13, pp. 47–48 (1965)	Grothendieck 1966; André 2004; Bost–Charles 2016	elliptic case statable
Homotopy hypothesis	134-2, p. 42: letter to Breen, 5 Feb. 1975	*Pursuing Stacks* (1983); Maltsiniotis 2010	partly
Section conjecture	149, p. 64 (undated)	letter to Faltings, 27 June 1983 (1997)	no étale π_1
Galois action and $\widehat{GT}$	119 (*Esquisse*, 1984); 142 ([from 1978]); 147	*Esquisse* (1997); Drinfeld 1990; Ihara 1990	partly
Generalised Hodge conjecture	12, pp. 2–3 (1964–[about 1971])	Grothendieck 1969 (*Topology* 8)	no Hodge structures on cohomology
p-curvature conjecture	93, pp. 193–197 ([1972]–1981)	Katz 1970 and 1972	partly

Table 1: The seven conjectures. None of them is in *Formal Conjectures* as of 17 October 2026 (code search on the repository).

2.1 The standard conjectures (1965)

On 27 August 1965 a letter to Serre states the standard conjectures, among them the positivity conjecture of Hodge type (shelfmark 14, p. 44); the letter is printed in the *Correspondance Grothendieck–Serre*. Folder 16 (1966–67) works out their algebraic side: Lefschetz operators, the pseudo-inverse Λ, Poincaré–Hodge algebras. Part of that algebra is proved in the Lean appendix (the normalisation of $\mathfrak{sl}_2$ and the pseudo-inverse, folder 16), and the register records one statement of the folder as possibly unpublished: the Lefschetz-type conjecture follows from $D(X)$ and the existence of *some* algebraic isomorphism $H^{2n-i} \to H^i$, not necessarily inverse to L^{n-i} (unsearched).

Folder 50 (dated by the inventory from 1965 on) sets out axioms for the endomorphisms of an abelian variety: a polarising class, the Rosati involution, an intersection form, a trace formula, and the positivity $\operatorname{tr}(\alpha'\alpha) > 0$ that Weil's proof of the Riemann hypothesis for curves rests on. The Lean appendix checks that $\nu(\alpha + n)$ is a monic polynomial of degree $2r$, and finds two counterexamples, both the edition's: the trace formula fails unless the intersection form is symmetric, which the axioms do not say; and the Rosati positivity does not follow from the other axioms (a model in a quaternion algebra), so it has to be assumed, as the Hodge-type conjecture has to be in general.

Status. Open in general: the Hodge-type conjecture holds in characteristic 0 by the Hodge–Riemann relations; the Lefschetz-type conjecture is known in dimension at most 2 and for abelian varieties.

2.2 The period conjecture (1965)

Folder 13, "Groupe de Galois motivique" (1965), asks on pp. 47–48 whether, over an algebraic extension k of $\mathbb{Q}$, the functor sending a motive to its de Rham realisation, its Betti realisation

and the comparison isomorphism between them is fully faithful. It reduces the question to $k = \overline{\mathbb{Q}}$ and recasts it as $V^G = V \cap g(\overline{V})$ for the motivic Galois group, noting that the rank-one (Tate) case works. This is the de Rham–Betti form of what is now called Grothendieck's period conjecture (register: *matched*, Bost–Charles 2016, Conj. 2.12).

Status. Open. The elliptic case can be stated with Mathlib's `PeriodPair` and `weierstrassP`: for a lattice with algebraic invariants and no complex multiplication, the periods and quasi-periods $\omega_1, \omega_2, \eta_1, \eta_2$ should have transcendence degree 4; with complex multiplication the degree is 2 (Chudnovsky, 1976).

2.3 The homotopy hypothesis (1975)

On 5 February 1975, the postscript of a letter to Lawrence Breen sets out the "yoga" by which an n-category in groupoids, up to n-equivalence, "est essentiellement la même chose" as a semi-simplicial set up to homotopy whose π_i are neglected for $i \geq n+1$ (shelfmark 134-2, p. 42). It is the homotopy hypothesis in its first form; *Pursuing Stacks* takes it up in 1983 for non-strict ∞-groupoids. The fonds also holds the handwritten French original of the third letter to Breen (17–19 July 1975, shelfmark 134-2, pp. 145–184), of which the folder otherwise holds only English versions, one of them R. Brown's translation.

Status. Depends on the model: false for strict n-groupoids from $n = 3$ (Simpson), regarded as open for Grothendieck's own globular ∞-groupoids in Maltsiniotis's formulation. An OpenAI preprint dated 24 September 2026, not peer-reviewed, announces a proof for every Grothendieck coherator in the Ara–Henry convention, together with Henry's pushout conjecture on elementary expansions (OpenAI, "The Grothendieck homotopy hypothesis via elementary expansions"). Its Lean companion (`lean/OAI/CategoryTheory/Globular` in the same repository, Apache 2.0: 58 files, about 12 750 lines, no `sorry`) defines globular theories, coherators and their models, and proves the elementary-expansion theorem; it does not state the comparison with spaces, which is the homotopy hypothesis itself.

2.4 The section conjecture

An undated page of folder 149 concludes "Nos conjectures impliquent donc qu'il y a exactement deux sortes de telles sections", those of rational points and those of points at infinity (shelfmark 149, p. 64). This is the section conjecture in its affine form, usually cited from the letter to Faltings of 27 June 1983.

Status. Open; injectivity is known, and so are a birational p-adic variant (Koenigsmann) and the real analogue.

2.5 The Galois action on Teichmüller towers

The *Esquisse d'un programme* (1984; folder 119) is at the origin of the group Drinfeld named after Grothendieck and Teichmüller. Folders 142, "Action de Galois sur Teichmüller" ([from 1978]), and 147, "Structure à l'infini des $M_{g,\nu}$", work on the same subject. Folder 142 lists necessary conditions for an automorphism of the profinite fundamental groupoid of a curve to come from $\mathrm{Gal}(\overline{K}/K)$ (pp. 94–101; register: *candidate*).

Status. Ihara's question — is $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \widehat{GT}$ an isomorphism? — is open; the map is injective by Belyi's theorem.

2.6 The generalised Hodge conjecture (1964–about 1971)

Folder 12, "Généralités (sous-groupes de Galois motiviques)", defines on pp. 2–3 the filtration of $H^i(\overline{X}, \mathbb{Q}_\ell)$ by the kernels of restriction to open sets whose complement has codimension $\geq j$ — the coniveau filtration — and expects $\mathrm{Filt}^j H^i$ to be the realisation of the largest sub-motive of $h^i(X)$ of level $\leq i - 2j$; granted resolution and the standard conjectures, its dimensions would

then be independent of ℓ. In the reading's words, this is the ℓ-adic form of what Grothendieck stated in 1969 as the corrected, generalised Hodge conjecture, after showing that Hodge's own general conjecture of 1950 is false.

Status. Open.

2.7 The p-curvature conjecture (1970s)

In folder 93, "Connexions de Gauss–Manin et équations de Picard–Fuchs" ([1972]–1981), pp. 193–197: for S integral, of finite type and dominant over $\operatorname{Spec}\mathbb{Z}$, and a vector bundle with an integrable connection on a smooth X/S, finite monodromy at the generic point implies vanishing p-curvature at almost all closed points; the page asks "la réciproque est-elle vraie ?", with rank-one and torsor variants. This is the Grothendieck–Katz p-curvature conjecture.

Status. Open; proved by Katz for Gauss–Manin connections (1972), and for connections with solvable monodromy, which include rank one (Chudnovsky for curves; Bost 2001; André 2004).

3 Questions posed in the notes, with no printed source

The register also holds questions that the notes raise and that the project has not found in print. They cannot go into *Formal Conjectures* as they stand, since that repository asks for a published source; they are listed so that a reader who knows the literature can say whether they are known.

- **The 27 lines around a triangle** (folder 69, "Graphes cubiques", [from 1976], pp. 7–8, 14–15, 25–26, 113; unsearched). The intersection graph of the 27 lines with a marked triangle t is described as the data of a 3-element set, an $\mathbb{F}_2$-plane V, and a torsor under the Baer product of three pull-backs of the extension of V by $\mathbb{F}_2$ that splits over no subgroup of order 2 — the quaternion group Q_8; $\operatorname{Aut}(C,t)$ has order $1152 = |W(F_4)|$. The group is classical; the torsor reconstruction may not be. A general version for graphs whose edges lie in unique triangles (folder 74) is proved in Lean.
- **Cohomological dimension and imperfection** (folder 31, SGA 4 residues, [1963–1973], p. 84; candidate). The bound on the ℓ-cohomological dimension of affine schemes of mixed characteristic, with residue characteristic ℓ, would be governed by the degree of imperfection $\nu(k)$, $\ell^{\nu(k)} = [k : k^\ell]$, rather than by the dimension. The local form, for an affine open of $\operatorname{Spec} A[1/\ell]$ with A henselian excellent, holds in a sharper form (an equality): O. Gabber, F. Orgogozo, *Sur la p-dimension des corps*, Invent. Math. 174 (2008), arXiv:0706.0671, Th. 1.2 and 6.1. The global bound $\sup(\operatorname{cd}_\ell k(x), \nu(V(\ell))+3)$ was not found; the paper's only nearby shift is the "$+2$" of Prop. 5.4.5, a technical step, and B. Bhatt, A. Mathew, *The arc-topology*, arXiv:1807.04725, Rem. 7.4, records the global mixed-characteristic case as open.

Some questions of the notes are now settled, and serve as a check on the method: whether the monodromy of projective structures on a fixed Riemann surface is injective (folder 159; yes, Poincaré); whether sequential colimits of presheaves on a test category are homotopy colimits (folder 107; not in general, Milnor's $\lim^1$); whether stratifying cohomology is invariant under nilpotent immersions (folder 7; no, as the printed exposé says); and in folder 68, "Jeux de position" ([from 1978]), a statement left under a question mark — that winning positions are reached by finitely many backward steps — which needs transfinite iteration, $\omega + 1$ already for a position with infinitely many successors. In folder 146 a table of trivialising exponents of tangent line bundles on $M_{0,n}$ under the symmetric group, with question marks for $n = 5, 6$, is now computed: $L = \bigotimes L_i$ has order $n-1$ and L_i order $(n-1)(n-2)$, proved in Lean for $n = 4, 5, 6$

at the level of cocycles and for every n in a homogeneous model of $M_{0,n}$ (functions of weight on configurations of vectors of K^2). The groups are known — $\mathrm{Pic}[M_{0,n}/\mathfrak{S}_n] = \mathbb{Z}/2(n-1)$ for n even, S. Gorchinskiy, F. Viviani, Math. Z. 258 (2008) 319–331, arXiv:math/0609705, Th. 3.6, after A. Arsie, A. Vistoli, Compositio Math. 140 (2004), arXiv:math/0301008 — but the orders of these bundles were not found stated.

4 Relation to recent machine-written proofs

On 6 October 2026 OpenAI made public a repository of 719 manuscripts produced by an internal model. Four families announce proofs of statements from Grothendieck's programme or bearing his name: the homotopy hypothesis for weak globular ∞-groupoids; the section conjecture, local p-adic and over $\mathbb{Q}$ for the modular curves $X_0(N)$ and $X_1(N)$ of genus at least two; the Deligne–Drinfeld conjecture on the Grothendieck–Teichmüller Lie algebra; and the rational Hodge conjecture for abelian varieties with complex multiplication, hence the Hodge-type standard conjecture for abelian varieties in any characteristic. None has been peer-reviewed, and only two are partly formalised in Lean. This note does not assess them. The fonds supplies dated points to which a confirmed proof can be related, and it does not attribute those conjectures: Deligne–Drinfeld is Deligne's and Drinfeld's, the Hodge conjecture is not Grothendieck's, and the proved forms of the homotopy hypothesis come from later work.

Table 2 sets out, for the seven conjectures of Table 1, what the preprints of `openai/math` claim and what they leave aside, as read in their abstracts on 17 October 2026.

5 Formalisation

What exists. The Lean appendix proves 2 052 theorems from the readings, on a Mathlib master of October 2026; general lemmas from it are proposed to Mathlib as pull requests. One overlap with *Formal Conjectures* was found: a proof in folder 36 (SGA 7, [about 1967–1973], on invariant cycles in H^1 over a henselian trait) holds "at least modulo resolution of singularities", whose solved and open cases are stated in `Wikipedia/ResolutionOfSingularities.lean`; the appendix now cites it.

What we propose. In order of feasibility:

1. the period conjecture, elliptic case, as `research open`, with the CM case as `research solved`; it needs only the Weierstrass zeta function and the quasi-periods on top of Mathlib's `PeriodPair`;
2. a weak form of Ihara's question for the group $\widehat{GT}_0$ defined by Drinfeld's relations (I) and (II) on the profinite free group $\widehat{F}_2$;
3. the section conjecture and the standard conjectures as issues only, with precise informal statements, until étale fundamental groups and Weil cohomology exist in Mathlib.

Each would cite the published source of Table 1, with the fonds as context.

6 Caveats and how to cite

- **Unchecked readings.** The transcriptions are first passes by a language model. A few pages have been read against the facsimile by a person; none of the folders above has been checked whole. Quotations are given as transcribed; the facsimile is the authority.
- **No priority claims.** A conjecture appearing in a folder says where Grothendieck wrote it, not that he was first; dates in brackets are the archivists' inferences.

Conjecture	OpenAI preprint	Claimed	Not claimed	Lean in the repository
Standard conjectures	*The rational Hodge conjecture for CM abelian varieties* (30 Sept.; 6 Oct., 53 pp.)	the Hodge conjecture for complex CM abelian varieties; as consequences, the Tate conjecture for abelian varieties over finite fields and the Hodge-type standard conjecture for abelian varieties in any characteristic	the standard conjectures beyond abelian varieties; B and D in general	none found
Period conjecture	none	—	the whole conjecture	—
Homotopy hypothesis	*The Grothendieck homotopy hypothesis via elementary expansions* (24 Sept., 26 pp.)	the hypothesis for every Grothendieck coherator in the Ara–Henry convention; Henry's pushout conjecture	other models of weak ∞-groupoids	58 files, about 12 750 lines: the elementary-expansion theorem; the comparison with spaces is not stated
Section conjecture	*The p-adic section conjecture* (24 Sept.; 6 Oct., 24 pp.)	the local p-adic conjecture for curves of genus ≥ 2 over every finite extension of $\mathbb{Q}_p$; the global conjecture over number fields when the finite-cover descent locus equals the rational points, including $X_0(N)$ and $X_1(N)$ of genus ≥ 2 over $\mathbb{Q}$	the global conjecture in general; the affine (cuspidal) form of the fonds	none found
Generalised Hodge conjecture	the same preprint as the first row	the generalised Hodge conjecture for CM abelian varieties, as a consequence	the conjecture beyond CM abelian varieties	none found
p-curvature conjecture	none	—	the whole conjecture	—
Galois action and $\widehat{GT}$	*The Deligne–Drinfeld conjecture* (23 Sept., 41 pp.)	the rational Grothendieck–Teichmüller Lie algebra is free on one generator in each odd weight ≥ 3	Ihara's question, $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \widehat{GT}$ an isomorphism	21 files under `Algebra/Drinfeld`, not inspected

Table 2: Coverage of the seven conjectures by the `openai/math` preprints (`https://github.com/openai/math`, folder `preprints`), from their abstracts and repository listings, 17 October 2026. None is peer-reviewed. "None found" means no Lean directory matching the subject in the repository's tree.

- **Rights.** The fonds is under copyright; publishing transcriptions requires the heirs' authorisation, which is being sought. Statements of mathematics are cited here; the manuscripts are not reproduced.
- **AI use.** The transcriptions, readings, register, Lean proofs and this note were produced with Anthropic's Claude models under the author's direction; the author checked and takes responsibility for the content.

References

- Y. André, *Une introduction aux motifs*, Panoramas et Synthèses 17, SMF, 2004.
- J.-B. Bost and F. Charles, "Some remarks concerning the Grothendieck period conjecture", *J. reine angew. Math.* 714 (2016).
- P. Colmez and J.-P. Serre (eds.), *Correspondance Grothendieck–Serre*, SMF, 2001.
- V. G. Drinfeld, "On quasitriangular quasi-Hopf algebras and on a group that is closely connected with $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$", *Leningrad Math. J.* 2 (1991).

- Google DeepMind, *Formal Conjectures*, `https://github.com/google-deepmind/formal-conjectures`.
- A. Grothendieck, "On the de Rham cohomology of algebraic varieties", *Publ. Math. IHÉS* 29 (1966).
- A. Grothendieck, "Standard conjectures on algebraic cycles", in *Algebraic Geometry (Bombay Colloquium, 1968)*, OUP, 1969.
- A. Grothendieck, "Esquisse d'un programme" and letter to G. Faltings (27 June 1983), in L. Schneps and P. Lochak (eds.), *Geometric Galois Actions* 1, LMS Lecture Notes 242, CUP, 1997.
- A. Grothendieck, *À la poursuite des champs*, vol. I, ed. G. Maltsiniotis, Documents mathématiques 20, SMF, 2022.
- Y. Ihara, "Braids, Galois groups, and some arithmetic functions", *Proc. ICM Kyoto* (1990).
- S. Kleiman, "Algebraic cycles and the Weil conjectures", in *Dix exposés sur la cohomologie des schémas*, 1968; "The standard conjectures", *Proc. Symp. Pure Math.* 55 (1994).
- G. Maltsiniotis, "Grothendieck ∞-groupoids, and still another definition of ∞-categories", arXiv:1009.2331 (2010).
- OpenAI, "The Grothendieck homotopy hypothesis via elementary expansions", preprint, 24 September 2026, `https://github.com/openai/math/blob/main/preprints/The-Grothendieck-homotopy-hypothesis-via-elementary-expansions-September-24-2026/paper.pdf`.
- OpenAI, *math*: mathematical manuscripts and supporting proof artifacts produced by an internal OpenAI model, `https://github.com/openai/math`, 6 October 2026.
- J. Stix, *Rational Points and Arithmetic of Fundamental Groups*, LNM 2054, Springer, 2013.
- A. Grothendieck, "Hodge's general conjecture is false for trivial reasons", *Topology* 8 (1969).
- N. Katz, "Nilpotent connections and the monodromy theorem", *Publ. Math. IHÉS* 39 (1970); "Algebraic solutions of differential equations", *Invent. Math.* 18 (1972).
- Université de Montpellier, fonds Alexandre Grothendieck, `https://grothendieck.umontpellier.fr`.